Regression to the Mean— Instructor Notes

Computing Commons

Study before class

Day: Fun Fridays · Professional Minds · **Week:** Week 5

Exact source: Jordan Ellenberg · *How Not to Be Wrong* · “The Triumph of Mediocrity”

Essential question: Did the intervention cause improvement, or did an extreme result naturally move closer to average?

Read the exact canonical source named above and use the session README for its pinned path and commit. Review the most important concepts before class.

1. Regression to the Mean— frame the move

Purpose: Make the essential question and the bounded finish line visible.

Say: Recognize stable factors plus chance, and demand comparison before causal stories.

Ask: What would a careful person need to know before accepting the first answer?

Do: Give students the one-sentence problem and let them predict.

Watch for: A polished answer being treated as evidence.

Move on when: Students can state the question and one uncertainty.

2. Today's finish line

Purpose: Turn the lesson into an observable student move.

Say: Extreme selection predicts some movement toward average.

Ask: Which part is a claim, and which part is an observation?

Do: Point to the three columns and name the receipt.

Watch for: Students trying to complete every possible extension.

Move on when: Each student can name a bounded output.

3. The mental model

Purpose: Give students a usable model without flattening the source.

Say: Stable factors plus chance explain regression after extreme selection.

Ask: Where does this model stop being enough?

Do: Use the boundary card explicitly.

Watch for: A metaphor replacing the actual criteria or evidence.

Move on when: Students can explain the model in their own words.

4. Worked computing example

Purpose: Translate the idea into a safe, clearly labeled computing case.

Say: This application is instructor-created; it is not a claim from the primary source.

Ask: What would we inspect first?

Do: Have pairs mark the checkpoint and alternative explanation.

Watch for: Students copying the example instead of transferring the move.

Move on when: Pairs can defend one checkpoint.

5. Ask before the answer

Purpose: Fade the scaffold by eliciting a prediction before revealing the frame.

Say: Did the intervention cause improvement, or did an extreme result naturally move closer to average?

Ask: What else could explain the same result?

Do: Collect one claim, one reason, and one uncertainty.

Watch for: Debate over conclusions without shared evidence.

Move on when: The room has at least two plausible explanations or criteria.

6. Do this

Purpose: Rehearse the professional action, not just the vocabulary.

Say: Name selection, stable factor, chance, and comparison.

Ask: Which step makes the work easier to inspect?

Do: Time a short individual attempt, then a partner check.

Watch for: Scope creep or a hidden leap from observation to conclusion.

Move on when: A partner can reproduce the route from the receipt.

7. Source and limits

Purpose: Protect source fidelity and bounded claims.

Say: Comparison/control logic tests causal stories.

Ask: What does this not prove?

Do: Read the limitation card aloud and invite one counterexample.

Watch for: Turning a pattern into a universal law or causal claim.

Move on when: Students can state one support and one limit.

8. Recovery

Purpose: Make revision and help-seeking part of the method.

Say: A failed check is information about the route, not a verdict about identity.

Ask: What assumption should we inspect first?

Do: Model one bounded retry and preserve the original observation.

Watch for: Repeated retries that change several variables at once.

Move on when: Students name a smaller next attempt.

9. Receipt

Purpose: Close with durable evidence and a next action.

Say: Selection, baseline, repeat, comparison, causal limit.

Ask: How will another person know what you actually checked?

Do: Give students the four-part receipt frame.

Watch for: Narration without the observation or limit.

Move on when: Each student has a concise receipt.

10. Exit

Purpose: Transfer the move to the next professional situation.

Say: Add a comparison or limitation to one before/after claim.

Ask: Where could this habit prevent a bad decision or unclear claim?

Do: Collect the exit sentence or point students to the next route.

Watch for: A new slogan without a concrete application.

Move on when: Students name a context and a verification step.